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PROJECT TOPIC:Food Delivery Management System

Assignment 5: Database Creation and Table Implementation

DBMS Coursework | Relational Schema Implementation using DDL Commands

1. Database Initialization

The physical database schema for the **Food Delivery Management System** is instantiated using Data Definition Language (DDL) scripts. Database initialization sets up character encoding, collations, and schema boundaries.

```
 - 1. DATABASE CREATION AND CONTEXT SELECTION
CREATE DATABASE IF NOT EXISTS FoodDeliveryDB
CHARACTER SET utf8mb4
COLLATE utf8mb4_unicode_ci;
```

```
USE FoodDeliveryDB;
```

2. Core Parent Entity DDL Implementation

Parent entities are created first to establish candidate primary keys for foreign key referencing by dependent transaction entities.

```
-- 2. CUSTOMER TABLE
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15) NOT NULL UNIQUE,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Address TEXT NOT NULL,
    Created_At TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 3. RESTAURANT TABLE
CREATE TABLE Restaurant (
    Restaurant_ID INT PRIMARY KEY AUTO_INCREMENT,
    Restaurant_name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15) NOT NULL UNIQUE,
    Address TEXT NOT NULL,
    Rating DECIMAL(2,1) DEFAULT 0.0 CHECK (Rating >= 0.0 AND Rating <= 5.0)
);

-- 4. DELIVERY PARTNER TABLE
CREATE TABLE Delivery_Partner (
    Partner_ID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15) NOT NULL UNIQUE,
    Vehicle_type VARCHAR(30) NOT NULL,
    Partner_status VARCHAR(20) DEFAULT 'Available'
        CHECK (Partner_status IN ('Available', 'Busy', 'Offline'))
);
```

```
[DDL LOG] Database `FoodDeliveryDB` created. Parent tables (Customer, Restaurant, Delivery_Partner)
executed successfully.
```


3. Dependent Entity DDL & Foreign Key Constraints

```
-- 5. MENU ITEM TABLE
CREATE TABLE Menu_Item (
    Item_ID INT PRIMARY KEY AUTO_INCREMENT,
    Restaurant_ID INT NOT NULL,
    Item_name VARCHAR(100) NOT NULL,
    Category VARCHAR(50) NOT NULL,
    Price DECIMAL(8,2) NOT NULL CHECK (Price > 0),
    Availability BOOLEAN DEFAULT TRUE,
    FOREIGN KEY (Restaurant_ID) REFERENCES Restaurant(Restaurant_ID) ON DELETE CASCADE
);

-- 6. ORDER TABLE
CREATE TABLE Order_Table (
    Order_ID INT PRIMARY KEY AUTO_INCREMENT,
    Customer_ID INT NOT NULL,
    Restaurant_ID INT NOT NULL,
    Order_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
    Total_amount DECIMAL(10,2) NOT NULL DEFAULT 0.00 CHECK (Total_amount >= 0),
    Order_status VARCHAR(20) DEFAULT 'Pending'
        CHECK (Order_status IN ('Pending', 'Preparing', 'Out for Delivery', 'Delivered', 'Cancelled')),
    FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID) ON DELETE CASCADE,
    FOREIGN KEY (Restaurant_ID) REFERENCES Restaurant(Restaurant_ID) ON DELETE CASCADE
);

-- 7. ORDER ITEM TABLE (Junction Table)
CREATE TABLE Order_Item (
    Order_item_ID INT PRIMARY KEY AUTO_INCREMENT,
    Order_ID INT NOT NULL,
    Item_ID INT NOT NULL,
    Quantity INT NOT NULL CHECK (Quantity > 0),
    Unit_price DECIMAL(8,2) NOT NULL,
    FOREIGN KEY (Order_ID) REFERENCES Order_Table(Order_ID) ON DELETE CASCADE,
    FOREIGN KEY (Item_ID) REFERENCES Menu_Item(Item_ID) ON DELETE CASCADE
);

-- 8. PAYMENT TABLE
CREATE TABLE Payment (
    Payment_Id INT PRIMARY KEY AUTO_INCREMENT,
    Order_ID INT UNIQUE NOT NULL,
    Payment_method VARCHAR(30) NOT NULL CHECK (Payment_method IN ('UPI', 'Credit Card', 'Debit Card',
'Net Banking', 'Cash')),
    Payment_status VARCHAR(20) DEFAULT 'Pending' CHECK (Payment_status IN ('Pending', 'Successful',
'Failed', 'Refunded')),
    Payment_date DATETIME DEFAULT CURRENT_TIMESTAMP,
    Amount DECIMAL(10,2) NOT NULL,
    FOREIGN KEY (Order_ID) REFERENCES Order_Table(Order_ID) ON DELETE CASCADE
);

-- 9. DELIVERY TABLE
CREATE TABLE Delivery (
    Delivery_ID INT PRIMARY KEY AUTO_INCREMENT,
    Order_ID INT UNIQUE NOT NULL,
    Partner_ID INT,
    Pickup_Time DATETIME,
```

```
    Delivery_Time DATETIME,
    Delivery_Status VARCHAR(20) DEFAULT 'Assigned' CHECK (Delivery_Status IN ('Assigned', 'Picked Up',
'Delivered', 'Failed')),
    FOREIGN KEY (Order_ID) REFERENCES Order_Table(Order_ID) ON DELETE CASCADE,
    FOREIGN KEY (Partner_ID) REFERENCES Delivery_Partner(Partner_ID) ON DELETE SET NULL
);
```

4. Schema Summary & Relationship Matrix

Table Name	Primary Key	Foreign Keys & References	Key Constraints
Customer	Customer_ID	None	UNIQUE(Phone, Email)
Restaurant	Restaurant_ID	None	CHECK(Rating 0.0-5.0), UNIQUE(Phone)
Delivery_Partner	Partner_ID	None	CHECK(Partner_status), UNIQUE(Phone)
Menu_Item	Item_ID	Restaurant_ID → Restaurant	CHECK(Price > 0), ON DELETE CASCADE
Order_Table	Order_ID	Customer_ID, Restaurant_ID	CHECK(Order_status, Total_amount)
Order_Item	Order_item_ID	Order_ID, Item_ID	CHECK(Quantity > 0)
Payment	Payment_Id	Order_ID → Order_Table	UNIQUE(Order_ID), CHECK(Payment_method)
Delivery	Delivery_ID	Order_ID, Partner_ID	UNIQUE(Order_ID), ON DELETE SET NULL